

A PROJECT TITLE

ON

Wine Feature-Space Visualization Using PCA and t-SNE

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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Department Of Computer Science and Engineering

Name of Title: **Wine Feature-Space Visualization Using PCA and t-SNE**

1. Abstract:

Wine quality assessment is a challenging problem due to the interplay of multiple physicochemical properties such as acidity, alcohol concentration, density, and residual sugar. These attributes collectively determine the sensory experience of wine, yet their high dimensionality makes direct interpretation difficult. This project aims to address that challenge by applying dimensionality reduction techniques to the wine dataset, thereby simplifying the feature space and enabling clearer visualization of the relationships between wine characteristics and quality ratings.

Two widely used methods, Principal Component Analysis (PCA) and t-Distributed Stochastic Neighbour Embedding (t-SNE), are employed to reduce the dataset into lower-dimensional representations. PCA provides a linear transformation that identifies the most influential features, offering a structured view of how certain variables dominate wine quality. On the other hand, t-SNE captures complex non-linear dependencies, forming clusters that group wines with similar properties. Together, these approaches provide complementary insights: PCA highlights global variance patterns, while t-SNE emphasizes local similarities and differences. This dual perspective allows for a more comprehensive understanding of the dataset.

The results of this study demonstrate that dimensionality reduction is not only useful for visualization but also for uncovering meaningful patterns that support predictive modelling. Wines with comparable physicochemical profiles tend to cluster together, suggesting that feature-based grouping can serve as a foundation for machine learning models aimed at quality prediction. By bridging statistical techniques with practical applications, this project underscores the importance of visualization in machine learning. It shows how complex datasets can be simplified without losing critical information, ultimately contributing to more efficient, data-driven approaches in the food and beverage industry. The findings highlight the potential of machine learning to enhance traditional quality evaluation methods, offering valuable insights for both researchers and wine producers.

2. INTRODUCTION:

The evaluation of wine quality has long been a subject of interest in both academic research and the food industry. Traditional methods rely heavily on expert tasting panels, which, while valuable, are subjective and time-consuming. With the rise of data science, there is growing interest in using machine learning to analyse physicochemical properties of wine, such as acidity, alcohol content, and sugar levels, to provide

objective insights. These datasets are often high-dimensional, making it difficult to interpret relationships directly without specialized techniques.

This project investigates the use of dimensionality reduction methods to simplify and visualize the wine dataset. Specifically, Principal Component Analysis (PCA) and t-Distributed Stochastic Neighbour Embedding (t-SNE) are applied to uncover patterns and clusters that reflect wine quality. By reducing complexity, these methods enable clearer interpretation of the data and highlight how certain features influence quality ratings. The study demonstrates how machine learning can complement traditional evaluation approaches, offering a structured and data-driven perspective on wine analysis.

3. SOFTWARE REQUIREMENTS:

Operating System: Windows 10/11, macOS, or Linux (any modern operating system capable of running Python).

Programming Language: Python version 3.8 or higher.

Libraries and Packages:

NumPy – for numerical computations.

Pandas – for data pre-processing and handling.

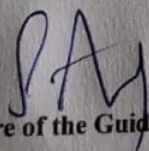
Matplotlib and Seaborn – for visualization.

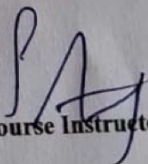
Scikit-learn – for machine learning algorithms including PCA.

sklearn.manifold (or OpenTSNE) – for t-SNE implementation.

Development Environment: Jupyter Notebook or Google Collab for interactive exploration and visualization. Alternatively, IDEs such as Visual Studio Code or PyCharm.

Version Control (Optional): Git or GitHub for managing source code and project versions.


Signature of the Guide


Course Instructor